SIMATS ENGINEERING

ASSESSMENT TOOL 1-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO1: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.
(Bloom Level -BL2- BL4)

Assessment Tool Weightage: 20%

Code of Conduct:

I Rahima Banu . N - 192525009 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rahima Banu N

PSEUDOCODE WRITING TASK

Q: Variables and Data types Analysis:

Pseudocode:

```
START

Set num = 20
Set name = "Rahima"
Set status = TRUE

sum = num + 10
result = status AND FALSE

Display num
Display name
Display status
Display sum
Display result

STOP
```

OUTPUT:

20
Rahima
TRUE
30
FALSE.

Q2: Data Structures Implementation

PSEUDOCODE:

START
Create Vector = [10, 20, 30]
Create List = ["Rahima", 19, TRUE]
Create Matrix = [[1, 2], [3, 4]].

Display Vector [2]

Update Vector [2] = 25

Find sum of vector

Display Vector
Display List
Display Matrix

STOP.

OUTPUT:

Vector: 10 25 30

List:

Rahima
19
TRUE

Matrix:
1 2
3 4

Sum = 65.

...rix and Array Operations:

PSEUDOCODE:

START

Create Matrix A

Create Matrix B

Add A and B

Multiply A and B.

Create 3D array

Display addition

Display Multiplication

Display Array

STOP.

OUTPUT:

Matrix Addition:

6	8
10	12

Matrix Multiplication:

19	22
43	50

3D Array Created Successfully.

Q4: Data Frame Construction and Manipulation:

PSEUDOCODE:

START
Create Data Frame (Name, Marks)
Add Grade Column
Delete Grade Column
Modify Marks
Display Data Frame
Display Summary
STOP

OUTPUT:

Name	Marks
Rahima	95

Q5: Functions and Logical Operations

PSEUDOCODE:

START
Create Function Square(x)
return x x x
Call Square (5)
Create Logical Vector
Perform Logical AND
Display Result
STOP

OUTPUT:

Square = 25
Logical Vector TRUE
FALSE
TRUE.

Good
11/7/26.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

98/100